

Road Report AI

Week 9 | 4/16/26 - 4/23/26

Leader: Jose Rodas

Kade O'Bar Time Spent 10 hrs	Tasks completed: • Updated the frontend and backend to the latest project state and aligned the new site with the current model-backed API. • Improved the report flow so the frontend uses richer backend data like score details, advice, weather context, and component breakdowns. • Reworked backend scoring to reduce the score clustering issue and make report results vary more realistically across roads and conditions. • Added a real weather fallback path so reports can still use live environmental context when the primary weather source fails. • Redesigned the methodology page into a dedicated technical explanation of how scoring, inputs, and fallback logic work. Planned tasks for next week: • Final Presentation Prep • Final tweaks Any issues or challenges: • Just more keeping the frontend connected to the most recent model
Dennis Keithly Time Spent 12 Hours	Tasks completed: • Fully rebuilt the weather processing for csv pipeline with Open-Meteo data, replacing the NOAA data that was originally being used ◦ Some redundant and overwritten work was done at the beginning of the week but much of the code was reused • Generated new types of training data both with combined section and road names and without • Attempted mapping of road sections to latitude and longitude coordinates Planned tasks for next week: • Ensure that the road segment mapping is adequate and restructure data if it is not • Integrate weather api with the model prediction api • Clean up general csv code and merge with main • Update SSoT with new csv pipeline information • Begin testing the deployed website and looking for bugs • If time permits begin looking to caching model output Any issues or challenges:

	• Fully rebuilt the csv weather processing pipeline ◦ Need to redo preprocessing and downloaded ◦ Needed some overhaul of generating negative samples • Very few simple solutions to getting the center of a certain road ◦ One solution is to download gigabytes of data ◦ Another is call large amounts of data through API though that comes with attempting to process dozens of different highway name configurations ◦ The solution I used was just averaging the coordinates of all crashes, but that may not be very accurate
Jose Rodas 7 hours	Tasks completed: • Pulled down the latest changes from teammates and ran the full app locally to spot-check the new scoring updates and weather fallback path before they went out. • Did some local fine-tuning on the model side, mostly playing with hyperparameters and threshold values to see how the score distribution moved. • Tested a bunch of edge cases on the new methodology page and report flow • Started putting together a rough outline for the final presentation and demo flow. Planned tasks for next week: • Work on slides and demo script for the final presentation. • Help with end-to-end testing on the deployed site • Keep tracking down any small bugs from edge case testing Any issues or challenges: • Final exams and other classes ate up most of the week • Trying to fine-tune the model locally without conflicting with other's work
Ben Stidham Time Spent 10 Hrs	Tasks completed: • Got the map view integrated with the back end, as to highlight and color-code nearby roads based risk scores • Continued integration between model and other components, including weather and crash reports Planned tasks for next week: • Bug and final integration testing and fixes • Backend model/road segment improvements Any issues or challenges: • Other projects and exams made me much busier
Christopher Chaiban	Tasks completed: • Updating the SSoT with current changes to project • Updating and adjusting deployment

Time Spent 7 hours	Planned tasks for next week: • Preparation for demo day • Any needed tweaks with the project • Assist with anything that requires immediate attention Any issues or challenges: • Busy weeks with other projects and final exams
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Total Time Spent: 46 Hours

Summary:

Solid week of integration work heading into the home stretch. Kade tightened up the connection between the new frontend and the model-backed API, including a richer report flow with weather context and a redesigned methodology page. Dennis fully rebuilt the csv weather pipeline on Open-Meteo and started mapping road sections to coordinates, which is going to set us up well for the location-aware predictions we've been wanting. Ben got the map view tied into the backend so nearby roads are color-coded by risk score, and Christopher kept the SSoT and deployment in sync with everything that's been moving. On my end I stayed in the background this week with local fine-tuning, edge case testing, and PR reviews, plus starting to put together a plan for the final presentation. Next week is mostly demo prep, end-to-end testing on the deployed site, and cleaning up any last bugs before we present.